SOP-103 — Mechanical Seal Inspection and Replacement - Process Pumps

Plant Site A · Rev 2 · Issued 28/11/2024 · Approved by Plant Engineering

1. Purpose

This procedure defines the approved method for inspecting and replacing cartridge mechanical seals on the horizontal centrifugal process pumps at Plant Site A. Seal leakage is the single most frequent corrective maintenance driver on the pump fleet, and disciplined seal work directly reduces repeat failures and unplanned downtime.

2. Scope

Applies to cartridge-type mechanical seals fitted to the process transfer pumps, including Pump 101A and its sister units, and to the utilities cooling water pumps. Gland packed auxiliary pumps are excluded. This procedure covers replacement in the field with the pump casing in place; full overhauls are performed in the workshop under separate instructions.

3. References

SOP-118 Equipment Isolation and Lock-Out Tag-Out.

SOP-101 Centrifugal Pump Startup and Shutdown Procedure.

Seal manufacturer installation drawing for the fitted seal model and size.

API 682 guidance on seal flush plans, held in the technical library.

4. Safety Precautions

The pump shall be isolated, drained, depressurised and tagged in accordance with SOP-118 before the coupling guard is removed. Verify zero energy at the local start station.

Process residues may be present in the seal chamber. Wear the gloves and face shield specified in the area PPE matrix while breaking containment.

Support the removed seal cartridge; do not allow it to hang on the shaft sleeve, as the faces crack easily under shock.

5. Tools and Materials

Have available before breaking containment: the replacement cartridge seal checked against the seal drawing for model and size, a dial indicator with magnetic base, feeler gauges, a calibrated torque wrench covering the gland nut range, new sleeve and gland gaskets, approved solvent and lint-free rags, fine emery cloth, laser or dial alignment equipment, and a containment tray sized for the casing volume.

Confirm the correct work order and pump tag at the machine before starting. The A, B and C units of a pump set are visually identical, and seal work on the wrong unit of a running set is a recurring industry incident.

6. Procedure

6.1 Confirm isolation per SOP-118 and verify the casing is drained and vented. Remove the coupling spacer and the coupling guard, and check shaft end float against the record card.

6.2 Before disturbing the seal, inspect and photograph the as-found condition. Note the leakage path: face leakage, sleeve gasket leakage and gland gasket leakage have different causes and the distinction matters for the failure record.

6.3 Where this is the second seal failure on the same tag inside twelve months, involve the reliability engineer before fitting the new seal. Fitting another seal into an uncorrected fault condition wastes both the seal and the outage.

6.4 Measure and record seal chamber run-out with a dial indicator. Run-out above 0.05 mm total indicator reading must be corrected before a new seal is fitted, or the new seal will fail early for the same reason as the old one.

6.5 Remove the seal gland nuts, back off the cartridge setting clips, and withdraw the seal cartridge along the shaft sleeve. Bag and tag the old seal for failure examination.

6.6 Inspect the shaft sleeve under the seal for fretting, scoring or corrosion. Polish minor marks with fine emery; replace the sleeve if damage can be felt with a fingernail.

6.7 Clean the seal chamber thoroughly. Confirm the flush port is clear by blowing instrument air through the line, and inspect the flush piping for blockage or crushed tubing. Check the flush orifice size against the piping specification while the line is open, and verify the flush flow direction matches the seal plan arrow on the drawing.

6.8 Fit the new cartridge seal square to the shaft, torque the gland nuts evenly in a cross pattern to the value on the seal drawing, and only then release the setting clips.

6.9 Rotate the shaft by hand through two full turns after fitting the seal and before refitting the coupling. Any rub or tight spot indicates the seal is cocked or a gasket is displaced, and shall be corrected before the guard is refitted.

6.10 Refit the coupling, verify alignment within 0.05 mm offset and 0.05 mm per 100 mm angularity, and refit the guard.

6.11 Restore the flush plan: open flush line valves, refill the seal pot where fitted, and vent the seal chamber through the gland vent connection until liquid appears.

6.12 Return the pump to service per SOP-101. Observe the seal for the first thirty minutes of operation; a correctly installed seal shows no visible leakage after an initial weep during the first minutes of running.

7. Acceptance Criteria

The replacement is accepted when the pump runs at normal duty with zero visible seal leakage, seal flush flow is confirmed, and bearing vibration is unchanged from the pre-work baseline. The old seal condition and the measured run-out shall be recorded on the work order before it is closed.

8. Records

Complete the work order with the failure code, the as-found seal condition, run-out readings, the new seal serial number and the downtime hours. Photographs shall be attached to the equipment history file for the pump tag. Seal failures achieving less than twelve months of running life shall additionally be reported to the reliability engineer so that repeat failure patterns on the same tag are visible and investigated.